

Stage 2: Practical Build Assessment

Self QA Checklist:

Happy Path:

Vehicle Search:

- Searching by a valid plate number returns matching vehicles.
- Clearing the search field restores the full vehicle list.
- Search continues to work correctly alongside status filtering.

Status Filtering:

- Selecting each status filter displays only vehicles matching that status.
- Selecting "All" displays the complete vehicle list.
- Filtering works correctly when combined with plate search.

Vehicle Details Panel:

- Clicking a vehicle row opens the details panel.
- The selected vehicle's information is displayed correctly.
- Vehicle details update when a different row is selected.
- No page navigation is required to inspect vehicle information.

CSV Export:

- Export button generates a downloadable CSV file.
- Export contains only currently filtered vehicles.
- Exported records match the visible table data.
- CSV opens correctly in spreadsheet applications.

Status Highlighting:

- Active vehicles display the correct visual status styling.
- Parked vehicles display the correct visual status styling.
- Maintenance vehicles display the correct visual status styling.
- Decommissioned vehicles display the correct visual status styling.
- Status highlighting remains consistent after filtering and searching.
- Highlights change upon hovering cursor and selection.

Edge Case Validation:

Empty Search Results:

- Searching for a non-existent plate returns zero results without errors.
- Interface remains responsive when no vehicles match search criteria.

Export with No Matching Records:

- Export operation does not crash when filters produce zero results.
- Empty export behavior remains predictable and stable.

Repeated Vehicle Selection:

- Multiple row selections correctly update the details panel.
- No stale vehicle data remains visible after changing selections.

Large Dataset Handling:

- Application remains usable with the provided dataset and no notable slowdowns.

Automated Test Coverage:

Automated tests were added for the newly added features:

- Vehicle filtering behavior
- Vehicle selection behavior
- CSV export functionality

These tests were chosen because they represent core user workflows and provide the highest confidence relative to implementation effort. These tests passed and were confirmed with “npm test” and lie within “vehicle-list.spec.ts”.